

THERMOREGULATION OF THE JAPANESE MACAQUE LIVING IN A SNOWY MOUNTAIN AREA

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Abstract A field investigation was conducted to study the thermoregulatory responses in nine Japanese macaques (*Macaca fuscata*) living in a snowy mountain area, Jigoku-Dani (Hell Valley, about 1,000 m above sea level) in Shiga Heights in central Japan in late January 1975. At about the same time, a laboratory study was made on four Japanese macaques reared in mild climate in an outdoor cage in Inuyama City. In the Hell Valley (HV) monkeys, no significant change in metabolic rate was observed at T_a between -1.4 and 28.3°C , while the rectal temperature was maintained at normal level. In the cold environment, the skin temperatures of HV monkeys were significantly higher than those of the monkeys living indoors previously studied. Similar patterns of metabolic and thermal responses were observed in Inuyama monkeys living outdoors, but to a lesser degree. The hair on the back and abdomen in the HV monkeys was significantly longer than that of Inuyama monkeys living indoors. It is suggested that the thick fur of HV monkeys may account for, if not all, the thermoregulatory responses of the Japanese macaque in snowy mountain areas.

The Japanese macaque (*Macaca fuscata*), whose habitat reaches as far as the Shimokita Peninsula (41°N), is the most northerly species of monkey found in the world. Japanese macaques of the Shimokita Peninsula and Shiga Heights are capable of surviving in the cold snowy forests and mountains during winter (IZAWA and NISHIDA, 1963; WADA, 1964). Several studies on thermoregulation of this species have revealed good defense reactions against cold stress, particularly

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when compared with the Crab-eating macaque whose natural habitat is in the tropical zone (NAKAYAMA *et al.*, 1971, 1975b; TOKURA *et al.*, 1975, 1976). However, the Japanese macaques used in these works do not represent the monkey living in a natural climate. The monkeys in these studies were reared in a animal house where the room temperature was controlled at $25 \pm 3^\circ\text{C}$ for 3–24 months before the experiments. In order to observe the climatic adaptation of this animal, it was necessary to study thermoregulatory response in natural conditions, or at least under conditions outdoor.

In January of 1975, we conducted a field study on the thermoregulation of the Japanese macaques living in the snowy mountain area at Jigoku-Dani (Hell Valley) in Shiga Heights in the central Japan. At about the same time, we studied the thermoregulatory responses of the Japanese macaques living outdoors in Inuyama city under less severe climatic conditions. We found that the two groups of the Japanese macaques developed an ability to acclimatize to cold if compared with the ability developed by the monkey living indoors in a previous report. A preliminary account of some of this work was published previously (NAKAYAMA *et al.*, 1975a).

DESCRIPTION OF LOCALE AND MONKEYS

A total of about 200 Japanese macaques (*Macaca fuscata*) in three troops live in Shiga Heights in central Honshu (36.4°N and 138°E ; an altitude of 800–1,600 m). The snow usually begins to fall early December and does not melt until April; it is usually 2–3 m deep. During the winter the monkeys are exposed to temperatures as low as -20°C . Each troop has its own territory in the area and migration from one troop to another is rare. Nine male adult monkeys of one troop (Shiga A-troop), consisting of 87 monkeys living in Jigoku-Dani (Hell Valley) (about 1,000 m) were used in the present study. Since 1963, this troop has been given provisions throughout the year by the staff of the Monkey Observatory of Hell Valley which is studying the social behavior of the monkeys. Provisions consisted of soybeans, wheat and apples. Assuming the monkeys consume all the food provided, the average energy of a daily ration is estimated to be 0.7–1.0 kcal for each individual, including adults and young. At night the monkeys rest in trees in the valley and at 7:00–8:00 AM they come down to the forest near the Observatory at the bottom of valley. Food is given once a day at approximately 8:30 AM. During the day they spend their time in the forest, eating and sitting in a nearby hot spring. The experiments were carried out at the Hell Valley Observatory from January 20 to 28, 1975, when temperatures ranged from -8 to $+5^\circ\text{C}$.

At the same time, four adult Japanese macaques (three males and one female), reared outdoors in Inuyama City, were studied. They had been captured in Fukui Prefecture in central Japan and reared in a large 20 m² outdoor cage

Table 1. Vital statistics of nine Hell Valley monkeys and four Inuyama outdoor-living monkeys.

Subject	Ages (Yr.)	Sex	Body weight (kg)	Surface area (m ²)	Skinfold thickness (mm)					
					Abdomen	Upper arm	Thigh	Back	Neck	Chest
Hell Valley monkeys										
HV-1	3	♂	8.90	0.5282	5.2	2.7	4.6	6.3	12.4	—
HV-2	3	♂	8.10	0.4961	3.8	2.6	3.7	5.2	8.0	—
HV-3	3	♂	7.95	0.4900	9.2	4.4	3.6	6.5	9.3	—
HV-4	3	♂	9.60	0.5556	8.2	3.2	4.0	7.6	11.0	13.6
HV-5	5	♂	14.13	0.7189	7.6	2.1	3.2	5.5	9.2	7.4
HV-6	3	♂	9.00	0.5321	4.6	2.1	2.6	5.2	6.6	7.1
HV-7	3	♂	7.08	0.4535	4.9	2.5	2.8	5.8	8.4	5.8
HV-8	3	♂	6.80	0.4415	3.6	2.8	2.8	5.2	9.0	5.9
HV-9	3	♂	10.35	0.5841	7.9	3.0	4.0	6.6	8.4	9.8
Mean±SD					6.1±2.1	2.8±0.7	3.5±0.7	6.0±0.8	9.1±1.7	8.3±3.0
Inuyama outdoor-living monkeys										
OD-1	<3	♂	11.20	0.6157	10.2	—	—	7.8	—	4.8
OD-2	<3	♂	6.20	0.4152	9.5	—	—	11.5	—	6.5
OD-3	<3	♂	13.00	0.6800	3.5	—	—	6.8	—	6.0
OD-4	<3	♀	11.00	0.6085	5.0	—	—	10.3	—	4.8
Mean±SD					7.1±3.3			9.1±2.2	5.5±0.9	
Inuyama indoor-living monkeys (NAKAYAMA <i>et al.</i> , 1971)										
4 animals	>3	♂ 2 ♀ 2	9.58±2.25	0.5512±0.0728						

There is no significant differences in these values among three groups of animals.

with a concrete shelter for at least a year before the experiment. The winter climate in Inuyama is much less severe than that of Shiga Heights. The temperature in January averages 3.8°C with a minimum low of -1.0°C. The snow is only a few cm deep at most. The experiments with Inuyama monkeys were done at the Primate Research Institute of Kyoto University in Inuyama from January 7 to 10, 1975. The vital statistics of the monkeys, along with skinfold thickness data, are shown in Table 1.

METHODS

Each Hell Valley monkey was captured by a trap cage near the Observatory at 8:30 AM. After being lightly anesthetized with ketamine hydrochloride (2.5 mg/kg, i.m.), the animal was placed on a restraining chair and was allowed to recover from the anesthesia in a room heated to a T_a of 15–20°C for at least two hours before the experiment started. The experiment was conducted in an experimental room located in a space under a balcony of the Observatory. The space was enclosed by a tent and snow to form an experimental room of 4 m². The room was slightly ventilated by an electric fan drawing the outside air through several holes in the wall. The air movement over the monkeys was lower than 10 m/min. Air hoses and thermistor wires ran through the tent to an adjoining room where the experimental equipment was placed.

The lowest temperature of the experimental room was dependent on the outside air temperature which was usually 1–2°C lower than that in the room. The higher room temperature was obtained by means of an oil stove in the room. By adjusting the flame of the stove, it was possible to control the room temperature at a nearly constant level. To prevent direct radiation from the heat source, a wooden screen was placed between the animal and the stove. Four Inuyama outdoor-living monkeys were placed on a restraining chair on the day prior to the experiment. The experiment was performed in a microclimate chamber where the fluctuation of ambient temperature was controlled to within 1°C. Since the comparison of thermoregulatory responses of the monkeys living under different thermal conditions is important in both experiments, care was taken to make the experimental conditions as close as possible to those of experiments with Inuyama indoor-living monkeys in a previous study (NAKAYAMA *et al.*, 1971). The relative humidity was 40 to 60% and the air movement over the subject was lower than 10 m/min (practically still air). The experiments started at 11:00–12:00 AM and ended at 3:00–4:00 PM. No food was given to the monkeys in the morning of experimental day.

Measurements of oxygen consumption were made by an open circuit method. Fresh outdoor air was drawn through a hood placed over the head of monkey by means of an air pump at a rate of 20–25 L/min. The animal's expired gas, diluted by the air, was then pushed into a dry gas meter and collected in a Douglas bag

for 5 min every 15 min. The gas analysis was made for the O_2 content by means of a Beckman E_2 analyzer. The metabolic rate was calculated from the O_2 consumption, on an assumption that RQ is equal to 0.83 (NAKAYAMA *et al.*, 1971).

Rectal (T_{re}), skin and ambient (T_a) temperatures were taken by means of thermistor probes and read on a meter every 5 min. A rectal probe was inserted 8 cm into the rectum. Skin thermistor probes (1×0.5 cm) were glued with a liquid cement and taped to a locally shaved part of the chest (just above the nipple), the thigh (middle part on the lateral side) and the foot paw (near the base of first metatarsus). T_a was measured at the level of the animal's chest. Modifying the equation determined for the Japanese macaque by HORI *et al.* (1972), the mean skin temperature ($\bar{T}_s$) was calculated by means of the following equation:

$$\bar{T}_s = 0.448T_{\text{chest}} + 0.411T_{\text{leg}} + 0.141T_{\text{foot}}$$

The surface area was estimated from the formula previously determined (HORI *et al.*, 1972). Tissue conductance was calculated, as stated in detail previously (NAKAYAMA *et al.*, 1971), from heat production, heat storage and respiratory evaporative heat loss, which was assumed to be 3.5 W/m^2 in a wide range of T_a . Shivering was visually observed and was also checked by an EMG.

In order to evaluate the insulative role of the fur of monkeys, the length of hair of four subjects was measured. Immediately after the experiments, hairs were clipped from the interscapular area of the back and from the abdomen just above the navel, leaving 2 mm of hair from the base. Fifty clipped hairs were taken at random for measurements of lengths. The measured length plus 2 mm was regarded as the true length of the hair. In a previous work (NAKAYAMA *et al.*, 1971), we did not measure the lengths of hairs in the indoor-living monkeys. In this study, therefore, we measured them in an indoor-living monkey (TK-89) which was considered representative of the monkeys previously studied in terms of age, sex, size, native place, nutritional state and thermal history. Measurements for the indoor-living monkey were made in July in order to conform to the season when the previous experiment was performed.

RESULTS

Table 2 shows individual data of metabolic rate, conductance and body temperatures at various T_a for 9 Hell Valley (HV) monkeys and 4 Inuyama outdoor-living (OD) monkeys.

Heat production

Figure 1 shows two typical records of thermoregulatory responses of HV monkeys to changes in T_a . In Fig. 1A, monkey HV-9 showed a decreased metabolic rate with rise in T_a . This type of response was more or less observed also in monkeys HV-2 and HV-6. On the other hand, monkey HV-5 showed an

Table 2. Steady-state values of thermoregulatory responses in nine Hell Valley monkeys and four Inuyama outdoor-living monkeys. Values are means $\pm$ standard deviation, with number of observations in parenthesis.

Subject	T_a (°C)	$\bar{T}_{re}$ (°C)	T_{re} (°C)	Heat production (W/m ²)	Conductance (W/m ² ·°C)
Hell Valley Monkeys					
HV-1	0.5 $\pm$ 0.2 (27)	24.8 $\pm$ 0.9 (27)	39.0 $\pm$ 0.3 (27)	43.8 $\pm$ 5.5 (8)	4.4 $\pm$ 2.2 (8)
HV-2	-0.6 $\pm$ 0.3 (13)	26.2 $\pm$ 0.2 (13)	38.3 $\pm$ 0.1 (13)	66.8 $\pm$ 7.8 (5)	5.1 $\pm$ 0.7 (5)
	10.9 $\pm$ 0.6 (17)	29.2 $\pm$ 0.6 (17)	38.3 $\pm$ 0.1 (17)	57.5 $\pm$ 13.8 (6)	5.8 $\pm$ 1.3 (6)
HV-3	0.8 $\pm$ 0.2 (25)	26.3 $\pm$ 0.2 (25)	39.2 $\pm$ 0.3 (25)	60.4 $\pm$ 10.6 (9)	4.7 $\pm$ 0.9 (9)
	10.6 $\pm$ 0.4 (13)	29.9 $\pm$ 0.3 (13)	38.8 $\pm$ 0.1 (13)	55.4 $\pm$ 3.1 (5)	5.0 $\pm$ 0.9 (5)
	27.5 $\pm$ 1.3 (4)	33.3 $\pm$ 0.1 (4)	39.5 $\pm$ 0.0 (4)	58.0 $\pm$ 7.6 (5)	7.7 $\pm$ 1.6 (5)
HV-4	3.5 $\pm$ 0.5 (20)	26.7 $\pm$ 0.4 (20)	38.6 $\pm$ 0.0 (20)	48.4 $\pm$ 10.1 (5)	3.8 $\pm$ 0.9 (5)
	16.9 $\pm$ 1.2 (22)	28.0 $\pm$ 0.2 (22)	38.7 $\pm$ 0.1 (22)	50.6 $\pm$ 4.0 (7)	4.5 $\pm$ 0.6 (7)
HV-5	-1.4 $\pm$ 1.0 (37)	27.1 $\pm$ 0.4 (37)	39.0 $\pm$ 0.2 (37)	53.6 $\pm$ 3.9 (12)	4.6 $\pm$ 0.9 (12)
	8.5 $\pm$ 1.5 (13)	28.7 $\pm$ 0.1 (13)	38.9 $\pm$ 0.0 (13)	52.5 $\pm$ 5.7 (5)	4.8 $\pm$ 0.6 (5)
	23.6 $\pm$ 0.4 (10)	30.2 $\pm$ 0.4 (10)	39.1 $\pm$ 0.1 (10)	63.3 $\pm$ 2.8 (6)	6.6 $\pm$ 0.5 (6)
HV-6	2.2 $\pm$ 0.9 (11)	28.7 $\pm$ 0.3 (11)	39.0 $\pm$ 0.1 (11)	47.8 $\pm$ 7.4 (5)	4.5 $\pm$ 0.6 (5)
	10.0 $\pm$ 1.5 (22)	30.4 $\pm$ 0.3 (22)	39.3 $\pm$ 0.1 (22)	40.6 $\pm$ 6.9 (6)	4.7 $\pm$ 0.9 (6)
HV-7	4.2 $\pm$ 1.2 (10)	30.1 $\pm$ 0.1 (10)	39.2 $\pm$ 0.1 (10)	56.3 $\pm$ 11.6 (7)	5.8 $\pm$ 0.9 (7)
	11.3 $\pm$ 1.3 (13)	31.8 $\pm$ 0.5 (13)	39.2 $\pm$ 0.1 (13)	61.0 $\pm$ 10.8 (5)	7.8 $\pm$ 1.2 (5)
	19.1 $\pm$ 0.8 (6)	34.5 $\pm$ 0.2 (6)	39.3 $\pm$ 0.1 (6)	64.2 $\pm$ 3.0 (3)	12.8 $\pm$ 0.3 (3)
HV-8	3.4 $\pm$ 0.4 (28)	28.3 $\pm$ 0.6 (28)	39.2 $\pm$ 0.2 (28)	65.3 $\pm$ 10.8 (5)	5.4 $\pm$ 0.3 (5)
	10.9 $\pm$ 0.3 (2)	28.7 $\pm$ 0.0 (2)	39.0 $\pm$ 0.0 (2)	63.5 $\pm$ 6.6 (2)	5.5 $\pm$ 1.2 (2)
	28.3 $\pm$ 1.6 (10)	34.9 $\pm$ 0.7 (10)	39.6 $\pm$ 0.2 (10)	62.6 $\pm$ 8.1 (4)	14.0 $\pm$ 1.5 (4)
HV-9	-0.5 $\pm$ 0.3 (13)	25.3 $\pm$ 0.3 (13)	39.2 $\pm$ 0.1 (13)	54.4 $\pm$ 3.8 (5)	3.7 $\pm$ 0.3 (5)
	5.8 $\pm$ 0.8 (8)	30.2 $\pm$ 0.6 (8)	39.4 $\pm$ 0.1 (8)	37.0 $\pm$ 2.6 (5)	3.2 $\pm$ 0.6 (5)
Inuyama outdoor-living monkeys					
OD-1	5.1 $\pm$ 0.2 (25)	24.9 $\pm$ 0.7 (25)	37.1 $\pm$ 0.5 (25)	51.5 $\pm$ 4.2 (9)	4.0 $\pm$ 0.3 (9)
	16.3 $\pm$ 0.3 (18)	30.3 $\pm$ 0.1 (18)	37.5 $\pm$ 0.0 (18)	47.6 $\pm$ 4.3 (6)	5.9 $\pm$ 1.1 (6)
OD-2	5.0 $\pm$ 0.3 (19)	26.8 $\pm$ 0.2 (14)	38.3 $\pm$ 0.0 (14)	69.9 $\pm$ 9.6 (5)	5.9 $\pm$ 0.7 (5)
	15.8 $\pm$ 0.5 (13)	31.2 $\pm$ 0.5 (13)	38.6 $\pm$ 0.3 (13)	62.1 $\pm$ 4.2 (4)	9.3 $\pm$ 1.5 (4)
	25.3 $\pm$ 0.6 (7)	34.8 $\pm$ 0.1 (7)	38.6 $\pm$ 0.1 (7)	56.7 $\pm$ 3.2 (2)	13.7 $\pm$ 0.7 (2)
OD-3	5.0 $\pm$ 0.3 (8)	27.7 $\pm$ 0.2 (8)	39.9 $\pm$ 0.0 (8)	46.2 $\pm$ 2.7 (5)	3.6 $\pm$ 0.2 (5)
	15.3 $\pm$ 0.6 (11)	31.4 $\pm$ 0.2 (11)	39.1 $\pm$ 0.1 (11)	39.8 $\pm$ 7.2 (4)	5.6 $\pm$ 1.6 (4)
	25.4 $\pm$ 0.4 (8)	33.5 $\pm$ 0.3 (8)	39.6 $\pm$ 0.1 (8)	37.5 $\pm$ 4.8 (3)	5.6 $\pm$ 0.5 (3)
OD-4	5.1 $\pm$ 0.3 (11)	25.5 $\pm$ 0.4 (11)	38.6 $\pm$ 0.2 (11)	57.9 $\pm$ 7.7 (4)	4.6 $\pm$ 0.3 (4)
	15.2 $\pm$ 0.6 (5)	29.9 $\pm$ 0.2 (5)	39.1 $\pm$ 0.3 (5)	38.3 $\pm$ 3.5 (4)	4.6 $\pm$ 0.9 (4)
	24.4 $\pm$ 1.4 (11)	31.8 $\pm$ 0.7 (11)	38.9 $\pm$ 0.1 (11)	41.4 $\pm$ 0.6 (4)	6.1 $\pm$ 0.7 (4)

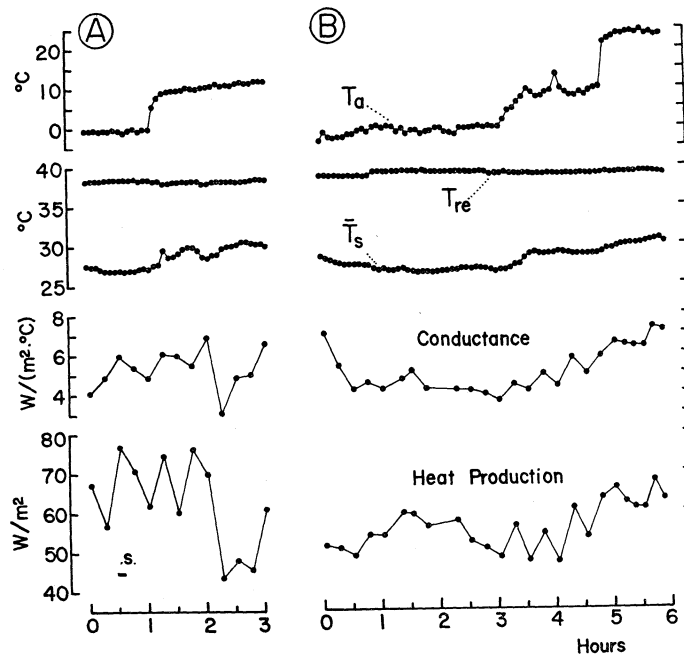


Fig. 1. Thermal and metabolic responses in two Hell Valley (HV) monkeys (monkey HV-9 in A and monkey HV-5 in B). T_a =ambient temperature. T_{re} =rectal temperature. $\bar{T}_s$ =mean skin temperature. s=shivering.

increased metabolic rate with the rise in T_a (Fig. 1B). The average metabolic rate of this monkey at T_a of 23.6°C was $63.3 \pm 1.3 \text{ W/m}^2$ (mean $\pm$ SE), which corresponds to 120.5% of that observed at T_a of 8.5°C . Monkey HV-7 also showed a similar type of response. The remaining three monkeys (HV-3, HV-4 and HV-8) did not change the metabolic rate with changes in T_a , the variation of which was within only 10%. Continuous and vigorous shivering was not observed in any of the monkeys. Four (HV-2, HV-3, HV-8, HV-9) of nine monkeys showed weak shivering occasionally when T_a fell to the near freezing point. When the steady state values of metabolic rate for individual are tentatively grouped into three according to T_a , the average metabolic rates are $55.2 \pm 2.8 \text{ W/m}^2$ at T_a between -1.4 and 5°C , $52.5 \pm 4.1 \text{ W/m}^2$ at T_a between 5.8 and 11.3°C and $59.7 \pm 2.8 \text{ W/m}^2$ at T_a between 16.9 and 28.3°C . There were no significant differences among these values ($P > 0.1$).

Figure 2 shows the thermoregulatory responses of two Inuyama outdoor-living monkeys (OD-3 and OD-4). At a T_a of 15.3°C , the metabolic rate of monkey OD-3 was between 34.3 and 50.3 W/m^2 , with an average of 39.8 ± 4.2 (SE), and then at a T_a of 25.4°C , it decreased slightly to 33.1 – 42.5 W/m^2 with an average value of $37.5 \pm 3.4 \text{ W/m}^2$. When T_a was subsequently lowered to 5°C , the monkey decreased the conductance to an average $3.6 \text{ W/(m}^2 \cdot ^\circ\text{C)}$ and increased the meta-

bolism to 42.6–48.7 W/m². The increase in metabolic rate during cold exposure was more conspicuous in the monkey OD-4 (Fig. 1 B), who showed shivering and an increased metabolism to 156.4% of that at T_a of 15.2°C. The other remaining two monkeys also showed increased metabolic rate at T_a of 5°C. The average increase in metabolism in the cold was 23.2% for monkey OD-2, while monkey OD-1 showed only an 8.2% increase. The Inuyama OD monkeys showed a small but definite increase in metabolism when exposed to a T_a of 5°C. The critical temperatures for metabolic increase seem to lie between 15 and 25°C for T_a and between 25 and 30°C for $\bar{T}_s$.

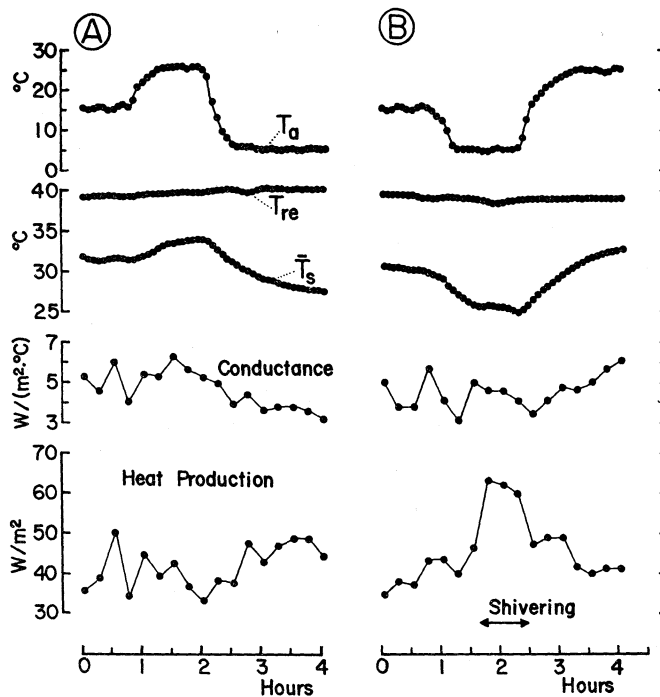


Fig. 2. Thermal and metabolic responses in two Inuyama outdoor-living monkeys (A for monkey OD-3 and B for OD-4).

In Figs. 3 and 4, the metabolic rates of HV monkeys are plotted against T_a and $\bar{T}_s$ in comparison with the mean values of the metabolic rates of Inuyama OD monkeys and of indoor-living monkeys previously studied (NAKAYAMA *et al.*, 1971). In the HV monkeys there was no change in metabolic rate at T_a between -1.4 and 28.3°C , and the critical ambient temperature for metabolic increase is suggested to lie below 0°C . These features of the metabolic responses in the HV monkeys are in good contrast to those of the indoor-living monkeys who showed more than two-fold increase in metabolic rate at T_a of 5°C with critical tempera-

tures of 22–29°C of T_a and of 32.5–35°C of $\bar{T}_s$. It is also apparent from Figs. 3 and 4 that the metabolic response of OD monkey is between the responses of HV monkey and the responses of indoor-living monkey.

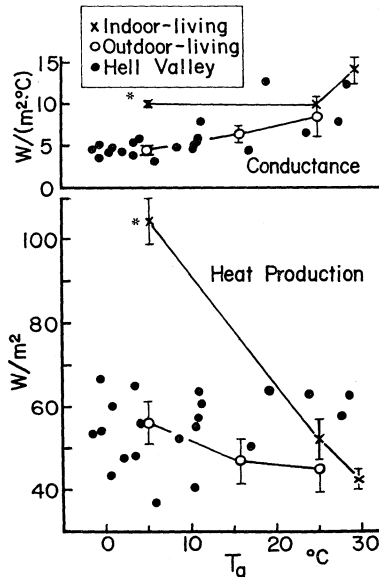


Fig. 3.

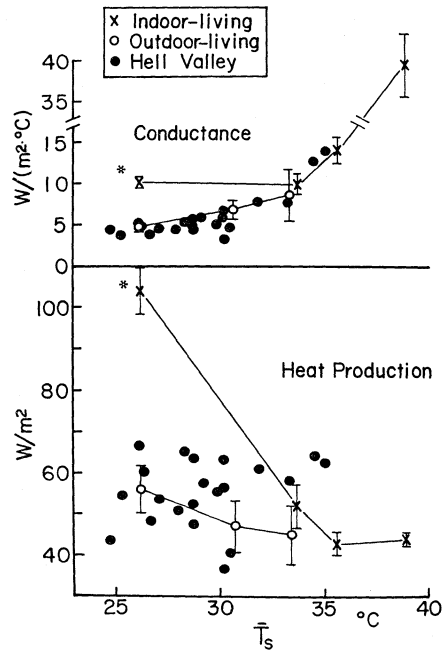


Fig. 4.

Fig. 3. Steady-state responses of HV monkeys to various ambient temperatures in comparison with the data of Inuyama outdoor-living (OD) monkeys and of indoor-living monkeys. A filled circle represents a mean steady-state value observed in one exposure for each HV monkey. Open circles show average values with standard errors of four Inuyama outdoor-living monkeys. Crosses denote average values with standard errors in the indoor-living monkeys previously studied (NAKAYAMA *et al.*, 1971). Asterisks indicate significant difference from the value of HV monkeys at T_a lower than 5°C, $P < 0.001$.

Fig. 4. Mean skin temperature versus heat production and conductance in HV monkeys, OD monkeys and the indoor-living monkeys. Symbols are the same as in the Fig. 3. Asterisk denotes the significant difference ($P < 0.001$) from the value at $\bar{T}_s$ between 24.8 and 27.1°C (mean $\pm$ SE, $26.1 \pm 0.9^\circ\text{C}$).

Another point to be questioned for HV monkeys is whether these monkeys in living snow maintain a high metabolic rate when compared with the indoor-living monkeys, since the high resting metabolism is one of the signs of non-shivering thermogenesis. Indeed, it has been reported that the cold-acclimated Japanese and rhesus macaques have rich brown adipose tissues and the cold-acclimated rhesus macaque maintains a high resting metabolism in a neutral

environment (CHAFFEE and ALLEN, 1973; ITOH and HIROSHIGE, 1967). The mean of the lowest values of metabolic rates for each of the nine HV monkeys was 50.5 ± 3.0 (SE) W/m², which is higher than those of the indoor-living monkeys (42.6 ± 2.9 W/m² (NAKAYAMA *et al.*, 1971) and 47.6 ± 5.0 W/m² (TOKURA *et al.*, 1975)) and of the OD monkeys (45.3 ± 7.1 W/m²). However, the difference was not significant ($P > 0.05$). Therefore, HV monkeys do not maintain higher metabolic rates than the other groups of monkeys as far as the present work is concerned.

Conductance

As shown in Fig. 3 and Table 2, tissue conductance was lower with decreasing T_a in most of the HV monkeys. To see the general tendency of conductance responses, the data is tentatively grouped into three according to T_a . The conductances are 9.1 ± 2.0 (SE) W/(m²·°C) at T_a of 16.9–28.3°C, 5.3 ± 0.6 at T_a of 5.8–11.3°C, and 4.7 ± 0.2 at T_a of 1.4–4.2°C. The conductance of HV monkeys at T_a of 5.8–11.3°C was significantly ($P < 0.005$) lower than that of indoor-living monkeys at T_a of 5.2°C.

Body temperatures

In HV monkeys, T_{re} were maintained within the range of 38.3–39.6°C at T_a of –1.4–28.3°C, in spite of no increase in metabolic rate in the cold (Fig. 5). The homeothermy in the cold was observed also in OD monkeys at T_a of 5–25°C. Exposure to cold led to a lowering of skin temperatures in the HV monkeys and the mean skin temperature ($\bar{T}_s$) was expressed as a function of T_a by the equation

$$\bar{T}_s = 0.25T_a + 26.90 \quad (r = 0.82). \quad (\text{Fig. 5})$$

The HV monkeys maintained a higher $\bar{T}_s$ in the cold when compared with indoor-living monkeys previously studied. In the HV monkeys, $\bar{T}_s$ was 28.85 ± 0.96 (SE) °C at T_a of 3.4–5.8°C, which is significantly higher ($P < 0.025$) than that of indoor-living monkeys (26.2 ± 0.48 °C at T_a of 4.5–6.0°C). This difference is more marked in the foot skin temperatures (T_{foot}). In a previous study, T_{foot} of indoor-living monkeys fell to 7°C after cold exposure (5°C) for two hours, while the average T_{foot} of HV monkey at T_a of 5°C was 16.9°C. The higher skin temperature was also found in OD monkeys, but to a lesser degree (Fig. 5).

Length of hairs

The average lengths of hairs on the back and abdomen in four HV monkeys and in an Inuyama indoor-living monkey (TK-89) are shown in Table 3. The hairs from both areas are significantly longer in HV monkeys than those in indoor-living monkeys. In both monkeys, the fur on back consists mainly of thick hairs with a few fine fibers. In indoor-living monkeys, the abdominal skin was sparsely covered with fine fibers, while in the HV monkeys, the abdominal and chest fur contained a considerable number of thick hairs. These fur features in the HV

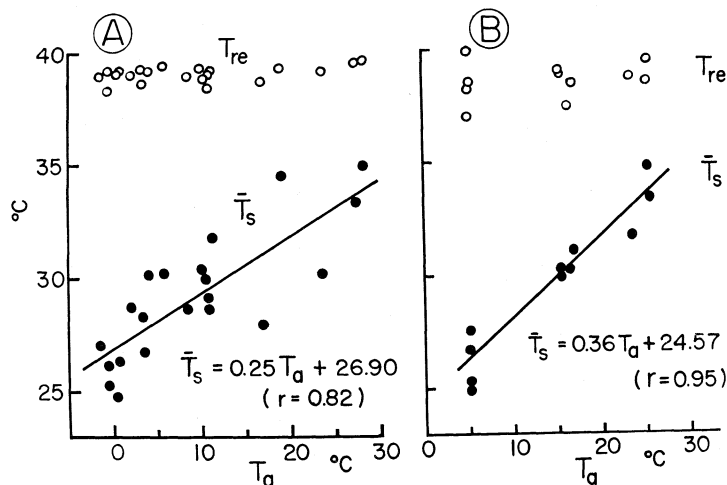


Fig. 5. Rectal (T_{re}) and mean skin temperature ($\bar{T}_s$) plotted against ambient temperature (T_a) in HV monkeys (A) and in the OD monkeys (B). Each point represents a mean of steady-state values observed in one exposure of one monkey.

Table 3. A comparison of length of hairs in four Hell Valley monkeys and an Inuyama indoor-living monkey (TK-89). Values (in mm) are means $\pm$ standard deviations. The number of observations is 50.

Subject	Back	Abdomen
HV-2	$7.9 \pm 0.5^*$	$7.2 \pm 1.7^*$
HV-5	$8.1 \pm 1.2^*$	$5.0 \pm 1.0^*$
HV-6	$7.8 \pm 1.1^*$	$4.9 \pm 1.0^*$
HV-7	$7.4 \pm 1.1^*$	$4.5 \pm 0.6^{**}$
TK-89	5.5 ± 0.5	4.1 ± 0.5

* $P < 0.005$. ** $P < 0.01$.

monkeys in winter suggest that they are provided with the great insulation in a cold environment, especially when the pilomotor activity goes into full operation. On the other hand, there is no evidence suggesting the increase in skinfold thickness in the HV monkeys.

DISCUSSION

It has been reported that at a T_a of 5°C, the indoor-living Japanese macaque maintained a core temperature at a normal level by 1.9 times (TOKURA *et al.*, 1975) and 2.5 times (NAKAYAMA *et al.*, 1971) increases in heat production, while in the present study, the Inuyama OD monkey in winter increased metabolism by only 1.3 times the resting value. In contrast, the HV monkey in the snow season

maintained a thermal balance without increase in heat production even at a T_a of -1.4°C . The critical ambient temperature for metabolic increase seems to be below the freezing point in the HV monkey. There is no evidence suggesting that the HV monkey maintains a higher resting metabolic rate than the other two groups of monkeys. Thus, it may be suggested that the constancy of deep body temperature in the HV monkey at T_a near 0°C is accomplished by an increase in insulation capacity, most probably by the marked increase in fur thickness, as was evident in the striking differences in length of hairs between the HV monkeys and the indoor-living monkey. The insulative role of pelage is well established in sheep (BIANKA, 1967; WEBSTER, 1966) and sled dogs (FOLK, 1974). According to WEBSTER (1966), the oxygen consumption at T_a of -10°C was greater in the sheep with a fleece depth of 15–25 mm than in the one with a fleece depth of more than 40 mm. There is a progressive decrease in critical temperature for metabolic increase in sheep as the fleece grows (BIANKA, 1967). Japanese macaques whose coats were clipped all over the body needed to increase their metabolic rate by 100% at T_a of 25°C and 53% at T_a of 5°C to maintain homeothermy, as compared with unclipped monkeys (TOKURA *et al.*, 1976). The fur of the wild Japanese macaque is sparse and short during summer and grows thick and long from autumn to winter. It is unknown whether the seasonal change of the fur thickness in the Japanese macaque is related to the environmental temperature or to the length of daylight time.

The contribution of subcutaneous fat tissues to the overall insulation in the HV monkey in winter is considered, if any, to be small. Although the body weight of the wild Japanese macaque is reported to reach maximum in November, the subcutaneous fat decreases drastically at the onset of the snow season. The body weight of male HV monkeys falls to almost minimum in February and does not recover until May when enough food is available (WADA, 1975). Two HV monkeys killed in accidents in April were almost void of subcutaneous and visceral fat in all over the body (WADA, 1964; WADA *et al.*, 1975). This indicates that the subcutaneous fat in HV monkeys has a nutritional role rather than an insulative role during winter. Although the HV monkey receives provisions and is supposed to be relatively well nourished, there was no evidence suggesting an increase in subcutaneous fat in the present work.

Thus, the type of cold acclimatization in the HV monkey does not seem to be a metabolic one, but rather an insulative one determined largely by the increase in fur thickness. In the snow season, the availability of food is extremely limited, qualitatively and quantitatively. The wild monkey depends on the barks of broad-leaf trees and winter buds of deciduous trees which have low caloric value (IZAWA and NISHIDA, 1963; WADA, 1964). Under these conditions, keeping the homeothermy by metabolic means alone must be difficult. CHAFFEE and ALLEN (1973) showed that the cold-acclimated (5°C) rhesus macaques maintained a higher resting metabolic rate and responded greatly to norepinephrine administra-

tion with increased oxygen consumption, indicating the presence of non-shivering thermogenesis in this animal. It would be interesting to study whether or not the HV monkey in the snow develops non-shivering thermogenesis which is more efficient in producing heat than shivering. This could not be studied in the present work in the field and remains to be worked out in the future. Although it has been reported that the cold-acclimated Japanese macaque has rich brown adipose tissues (ITO and HIROSHIGE, 1967), the presence of brown adipose tissue in the HV monkey in winter has not so far been determined (WADA *et al.*, 1975).

Another finding to be pointed out in the HV monkey is the high skin temperature in the cold, as compared with that of the indoor-living monkey. This is particularly conspicuous in the skin temperature of foot paw where the fur is sparse. During immersion of finger in 0°C water, the HV monkeys in the present study showed earlier vasodilatation at higher finger temperatures, as compared with the indoor-living Japanese macaque (OKADA, 1975; OKADA *et al.*, 1975). Even when the fingers of monkeys HV-6 and HV-9 were immersed in the snow (−4 to −6°C) for 30 min at T_a of 10°C, the finger skin temperature could be maintained at 7 to 12°C. These characteristics in the skin temperature and vasomotor responses in the HV monkey in winter indicates a tolerance for local cold stress that enables them to perform delicate finger manipulation in processes such as grooming.

It is interesting that these patterns of metabolic and skin temperature responses observed in the cold-acclimatized HV monkeys were also found, but to a lesser degree, in the Inuyama outdoor-living Japanese macaques in winter. Since the OD monkeys in the present work are similar to the indoor-living monkeys used in the previous study, in terms of age, size, nutritional state and the native place, the difference in thermoregulatory responses between these two groups of monkey may be ascribed to the difference of thermal conditions of breeding.

The features of cold acclimatization described above do not constitute a complete profile of thermoregulation in the HV monkey in winter. Another important regulation against cold is a behavioral one. Various kinds of thermoregulatory behaviors have been reported by ecologists (WADA, 1964, 1975). Huddling is frequently observed on the windy and snowy days. On the sunny days, they move to seek the sunny slopes of sheltered spots where they can avoid the wind and find winter buds for food easily. We also observed that in snow they prefer to take a peculiar posture when resting, both limbs are flexed and are compressed firmly on the ventral side of the body, thereby reducing the heat loss from the chest, abdomen and groin areas where the fur is sparser than that on the back. Sitting in a hot spring also seems to show thermoregulatory behavior. The water temperature in the pond was 38–41°C. According to our limited observations during the short stay there, the number of animals sitting in the spring and the staying time in spring seems to be related to the weather conditions. Further studies will be required to evaluate the thermoregulatory role of sitting in a hot

spring.

The HV monkey in the present work is, in a strict sense, not a wild monkey in snowy mountain areas. Their thermal environment is natural but their food availability was artificial. The other two troops living in the Shiga Heights are not provided with food and the problem of food availability in winter is more serious for them. Indeed, one of the troops is reported to reach as high as 1,600 m above sea level, where they are exposed to more severe conditions, in terms of temperature and food availability. However, the properties of the cold acclimatization of these truly wild snow monkeys do not suggest much difference from those of HV monkeys, at least, qualitatively.

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